Predicting Individual Medical Insurance Charges

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Project Contributions: Yu Wang contributes to the interaction model, Box-Cox, stepwise selection, cross-validation and region ANOVA; Yicheng Yang contributes to the age-linearity check, spline comparison, results and conclusion revision; Yujun Zhang contributes to Introduction, EDA and baseline OLS;

1. Introduction

The dataset analyzed in this project is drawn from the Kaggle dataset 'Medical Cost Personal Datasets'. This dataset contains 1338 observations with seven variables describing individual medical insurance beneficiaries in the United States. The response variable, charges, is the individual medical costs billed by health insurance. The six predictors include three numeric variables (age, bmi, children) and three categorical variables (sex, smoker, region). There are no missing values in the dataset.

The objective of this project is to apply the regression and diagnostic techniques developed in STAT 425 to explore and quantify how individual characteristics influence the annual insurance charges. By building and comparing several regression models, we seek to identify a final model with the strongest generalization performance.

We structure our analysis around three guiding research questions:

1. Beyond their individual effects, do any pairs of predictors interact in ways that meaningfully shape charges?
2. Do charges vary systematically across the four U.S. regions, and if so, does this variation persist after accounting for individual risk factors?
3. Is the effect of age on charges truly linear, or does cost accelerate at certain life stages?

2. Exploratory Data Analysis

2.1 Variable Types and Descriptive Statistics

The dataset contains three numeric predictors: age (integer, 18–64), bmi (continuous body mass index), and children (integer count of dependents, 0–5) and three categorical predictors: sex (female/male), smoker (yes/no), and region (northeast, northwest, southeast, southwest). The response variable charges is continuous and positive. Table 1 summarizes the numeric variables.

Variable	Mean	SD	Min	Median	Max
age	39.21	14.05	18.00	39.00	64.00
bmi	30.66	6.10	15.96	30.40	53.13
children	1.09	1.21	0.00	1.00	5.00
charges	13,270.42	12,110.01	1,121.87	9,382.03	63,770.43

Table 1. Descriptive statistics for numeric variables ($n = 1{,}338$).

The categorical variables are reasonably balanced: sex contains 662 females and 676 males; the four regions are roughly equal in size (northeast 324, northwest 325, southeast 364, southwest 325). The variable smoker, however, is heavily imbalanced, with 1,064 non-smokers and only 274 smokers, a ratio of approximately 4:1.

2.2 Visual Exploration

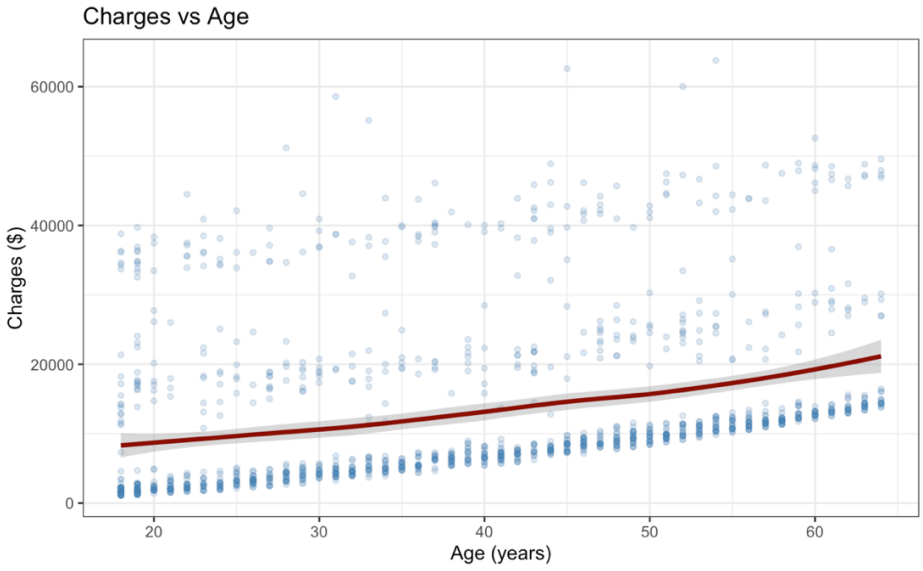


Figure 1: Charges increase broadly linearly with age. The curve shows mild curvature, which we will formally test in the following chapters by using polynomial regression and natural splines.

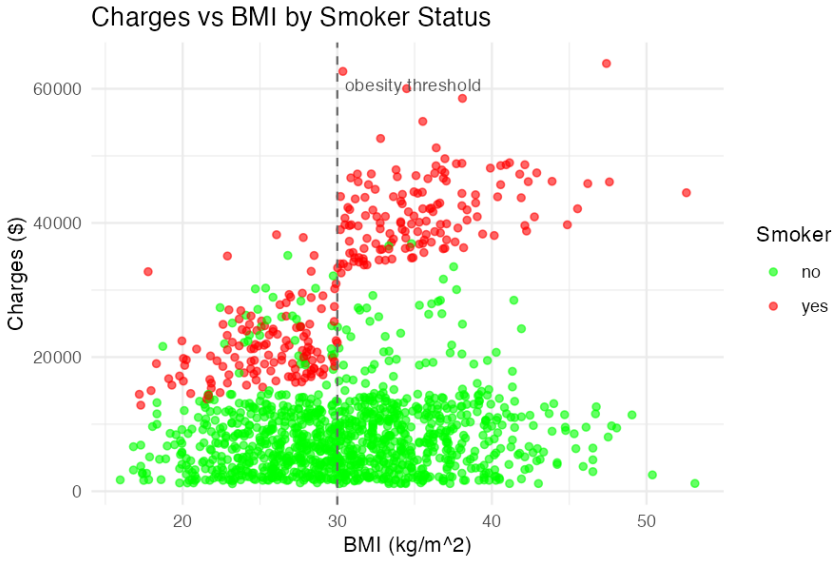


Figure 2: For non-smokers, BMI has little association with charges. For smokers, charges escalate sharply once BMI exceeds 30, suggesting a sign of multiplicative interaction.

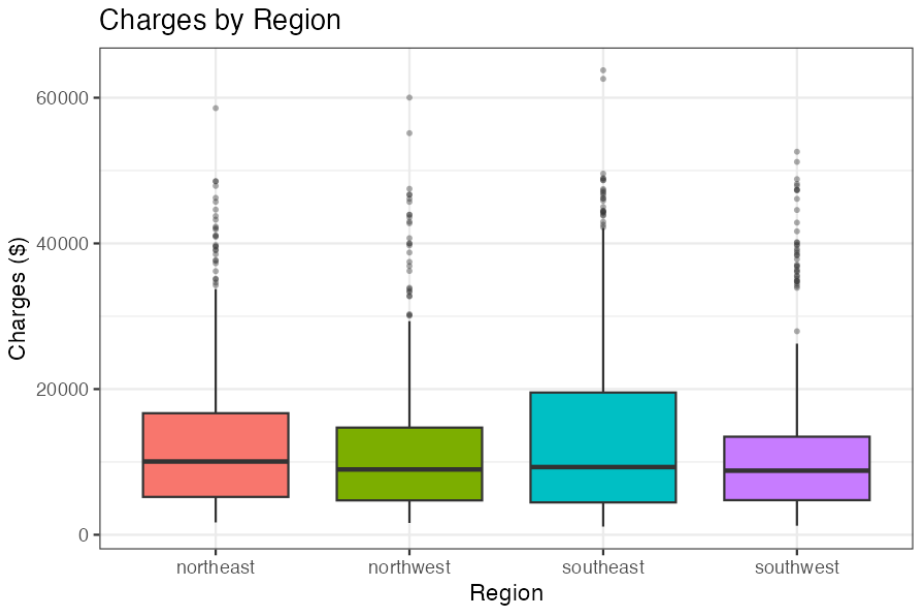


Figure 3: The four regions show similar charge distributions, raising the question of whether regional differences are truly absent or merely obscured by confounding risk factors.

2.3 EDA Implications for Modeling

Three patterns emerging from the visual exploration directly shape our modeling strategy.

Firstly, and most consequential, Figure 2 reveals that the effect of BMI on charges differs a lot across smoking groups: among non-smokers, BMI shows little association with charges, whereas among smokers, charges increase sharply once BMI exceeds the clinical obesity threshold of 30 kg/m^2. This interaction cannot be captured by additive terms alone, and directly motivates the inclusion of both a continuous bmi:smoker interaction and a discrete obese:smoker interaction in Model B.

Secondly, the near-identical regional boxplots in Figure 3 do not necessarily imply that region is irrelevant. The marginal similarity may reflect confounding: if high-risk individuals (smokers, high BMI) are unevenly distributed across regions, their dominant effect on charges could mask genuine regional variation. We will investigate this possibility by examining the region's partial effect after controlling for all other predictors.

Thirdly, the relationship between age and charges appears broadly linear. Rather than assuming linearity, we will formally test this assumption later by using orthogonal polynomials and a natural cubic spline.

3. Methodology

We split the data into training (n = 1,070) and test (n = 268) sets using seed 425 (an 80:20 split). All Model selection and diagnostics are performed on the training set; the test set is reserved exclusively for final out-of-sample comparison. We define the binary indicator **obese = 𝟙{bmi ≥ 30}**, where 30 is the WHO clinical obesity threshold. This cutoff is chosen a priori from medical convention, not learned from the data, so it does not introduce data-snooping bias.

3.1 Model A: Baseline OLS

We begin with the simplest possible specification: a multiple linear regression with all six predictors entered in their original form, without transformation or interaction.

$$\text{charges} = \beta_0 + \beta_{1\text{age}} + \beta_{2\text{bmi}} + \beta_{3\text{children}} + \beta_{4\text{sex}} + \beta_{5\text{smoker}} + \beta_{6\text{region}} + \varepsilon,\ \varepsilon \sim N(0, \sigma^2)$$

The model is fit by ordinary least squares on the 1,070 training observations, with the estimator $\hat{\beta} = (X^T X)^{-1} X^T y$. The training $R^\wedge 2 = 0.7555$ and adjusted $R^\wedge 2 = 0.7536$, with the overall F-test highly significant ($F = 409.70$ on 8 and 1,061 degrees of freedom, $p <$ 2.2e-16).

Collinearity: All predictor VIFs are well below 2, indicating there are no collinearity concerns.

Residual diagnostics. The formal tests and Figure 4 agree on what's wrong with Model A.

Constant variance fails: BP = 104.29 (p = 5.64×10^{-19}). The residuals-vs-fitted plot shows why: residuals split into two clear bands, non-smokers below and smokers above, separated by several thousand dollars. This isn't random heteroscedasticity, it's a structural variable missing from the model. The upward-sloping curve in the scale-location panel says the same thing.

Normality also fails (W = 0.898, p = 4.82×10^{-26}), with both tails of the Q-Q plot bending off the reference line.

Independence is fine, DW = 1.997 (p = 0.479), the only assumption Model A satisfies.

No single observation is distorting the fit: the Bonferroni outlier test flags only two points, the largest Cook's D is 0.025 (well below 1), and nothing in the residuals-vs-leverage plot crosses the Cook's contour. So the problem isn't a few weird points; it's a missing piece of structure that §3.2 will fix by adding the smoker-BMI interaction.

Test set performance: On the testing dataset, Model A achieves Test RMSE = 6,088.63 and Test MAE = 4,221.07. These benchmark values establish the baseline against which all subsequent models are compared.

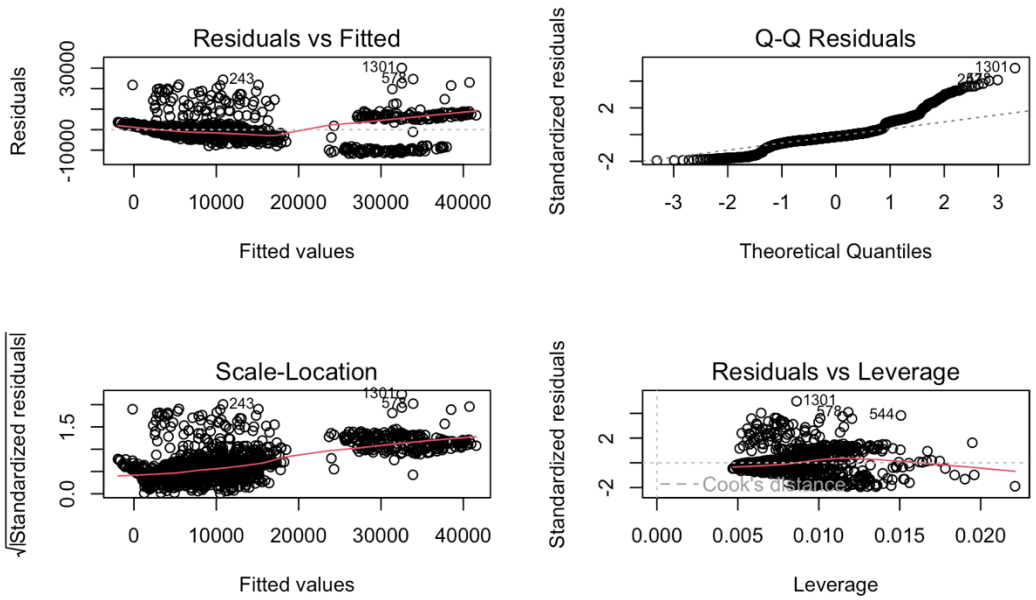


Figure 4: The four standard diagnostic plots.

3.2 Model B: Interaction Model

Model A assumes that all predictors act additively on charges. Research Question 1 asks whether this assumption is too restrictive: beyond their individual effects, do any pairs of predictors interact in ways that meaningfully shape medical charges? To answer this systematically, we added each two-way interaction to Model A one at a time and compared the augmented model against Model A using partial F-tests on the training set.

Interaction screening. Among all pairwise interactions, **bmi × smoker** is by far the dominant signal, with F = 591.00 and p < 2.2e-16. It increases adjusted R^2 by 8.81 percentage points, whereas the next strongest interaction, smoker × region, increases adjusted R^2 by only 0.31 percentage points. Sex × smoker is borderline after multiple-testing adjustment and contributes little predictive gain. Thus, although several interactions are statistically detectable, only bmi × smoker is interpretable and predictively meaningful.

Interaction	**df**	**F**	**p-value**	**BH-adjusted p**	**ΔAdj. R^2 (pp)**
bmi × smoker	1	591.00	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	+8.81
smoker × region	3	5.45	0.0010	0.0076	+0.31
sex × smoker	1	6.64	0.0101	0.0506	+0.13
age × smoker	1	3.49	0.0621	0.2328	+0.06

Table 2. Top pairwise interactions.

Model B is therefore

$$charges = \beta_0 + \beta_1 age + \beta_2 bmi + \beta_3 children + \beta_4 sex + \beta_5 smoker + \beta_6 region + \beta_7 obese + \beta_8 (bmi \times smoker) + \beta_9 (obese \times smoker) + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Joint significance and prediction. The partial F-test of Model A against Model B yields $F = 306.58$ on 3 and 1,058 degrees of freedom ($p < 2.2 \times 10^{-16}$), confirming that the three added terms jointly carry substantial signal beyond the additive baseline. Training R^2 rises from 0.7555 to 0.8692 (adjusted $R^2 = 0.8678$), and 5-fold CV RMSE drops from \$6,049.69 to \$4,415.54. Adding smoker × region on top of Model B yields no further signal (partial F $p = 0.5067$), so Model B is retained as specified. Held-out test-set comparison appears in Table 4.

Coefficient interpretation. For non-smokers, BMI has no detectable effect on charges. For smokers, each additional BMI unit adds roughly \$489 in annual charges beyond the non-smoker baseline, and crossing the obesity threshold contributes a further discrete jump of \$14,937, by far the largest and most precisely estimated term in the model. BMI matters, but it matters almost entirely through smoking status, a conditional structure that Model A cannot represent.

Table 3. BMI's effect on annual charges in Model B, decomposed by smoker status.

Effect	**Estimate (\$)**	**95 % CI (\$)**	**p-value**
BMI slope, non-smokers	13.56	[−69, 96]	0.747
Additional BMI slope, smokers	488.93	[312, 666]	6.88×10^{-8}
Discrete jump at BMI = 30, smokers	14,936.81	[12,685, 17,188]	$< 2 \times 10^{-16}$

Diagnostics. The Model B diagnostic plots in Figure 5 show that the two-band residual pattern from Model A has largely disappeared.

The Breusch-Pagan test confirms the change: BP = 5.38 ($p = 0.9115$), compared with Model A's $p = 5.64 \times 10^{-19}$. The apparent heteroscedasticity in Model A was therefore largely a symptom of omitted interaction structure rather than genuine variance heterogeneity.

Normality, however, is not repaired: the Q-Q plot still shows right-tail deviation and the Shapiro-Wilk test continues to reject normality ($W = 0.465$, $p = 1.22 \times 10^{-48}$), driven by a small cluster of high-cost observations. With $n = 1{,}070$ and no extreme leverage points, this localised tail departure does not undermine coefficient inference, and it motivates the response-transformation experiment in §3.3.

Test set performance.

Table 4. Held-out test-set performance (n = 268).

Model	RMSE ($)	Δ vs A	MAE ($)	Δ vs A
Model A (additive baseline)	6,088.63	—	4,221.07	—
Model B (bmi × smoker + obese × smoker)	4,351.99	−28.5 %	2,301.68	−45.5 %
Model B + smoker × region	4,368.10	−28.3 %	2,301.96	−45.5 %

Model B reduces test RMSE by 28.5 % and test MAE by 45.5 % relative to Model A, while adding smoker × region on top of Model B yields no further gain. Model B is therefore the predictive operating point of §3.2.

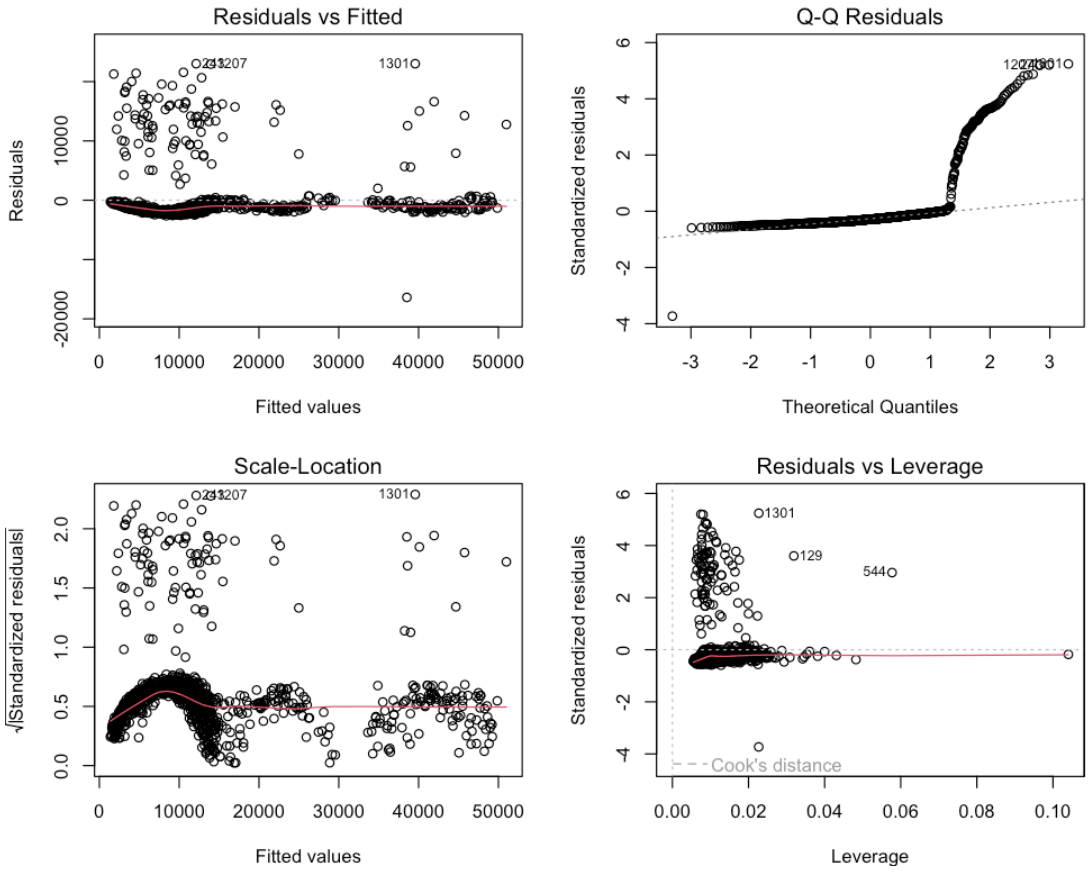


Figure 5: Diagnostic plots for Model B.

3.3 Model C: Box-Cox Transformation Experiment

Although Model B passes the Breusch-Pagan test, the Shapiro-Wilk test continues to reject normality, and the Q-Q plot shows pronounced right-tail deviation. In §3.3 we test whether a power transformation of the response can address this residual problem.

We apply the Box-Cox procedure to Model B's predictor structure to let the data identify the optimal transformation by maximum likelihood, obtaining $\hat{\lambda} = 0.33$ with 95% confidence interval [0.27, 0.38]. This interval excludes both 0 (log) and 1 (no transformation), indicating the data prefer a near-cube-root transformation over either conventional choice.

We back-transform predictions to the dollar scale using Duan's smearing correction (empirical factor 1.073) so that Model C's test errors are directly comparable to Models A and B.

Table 5. Comparison between Model B & Model C

Metric	Model B	Model C	Δ
Test RMSE	4,351.99	5,465.81	+25.6%
Test MAE	2,301.68	3,265.88	+41.9%
BP test (p)	0.9115	0.0079	reintroduced
SW test (W)	0.465	0.678	improved but still rejects

The Q-Q plots show that Model C does not improve the residual problem. Model B only has noticeable deviation in the upper-right tail, while Model C also distorts the middle part of the residuals. In other words, the Box-Cox transformation did not fix the tail issue and instead made the overall residual pattern worse.

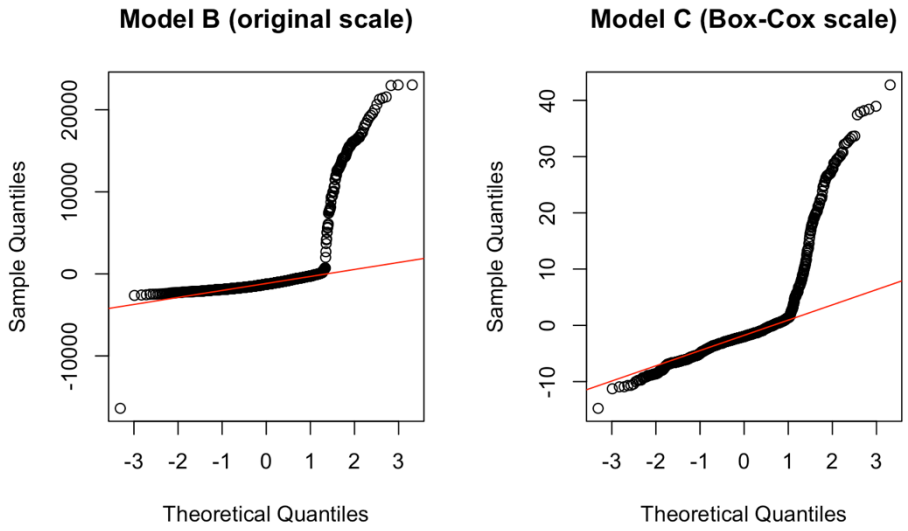


Figure 6: Q-Q plot comparison for Model B and Model C.

3.4 Stepwise Variable Selection

We apply stepwise selection in both directions using AIC and BIC as competing criteria. The two key interactions (smoker × bmi and smoker × obese) are forced into the lower bound of the search space, since they are theoretically justified by the EDA findings of Section 2.

AIC selection ($k = 2$) retains all variables; BIC selection ($k = \log n \approx 6.975$) drops region and sex, yielding Model B_step with 8 parameters, down from Model B's 12. A partial F test comparing Model B_step against Model B gives $F = 3.83$ on 4 and 1,058 degrees of freedom, $p = 0.0043$, formally significant at $\alpha = 0.05$, but the dropped terms account for only 1.4% of the residual sum of squares. Whether to adopt the BIC reduction is revisited in Section 4.

3.5 Five-Fold Cross Validation

We perform 5-fold cross-validation on the training set, comparing Models A, B, and Bstep. Because the obese indicator uses a fixed WHO threshold (BMI = 30) rather than a data-derived cutpoint, no leakage protection across folds is required; all errors are on the original dollar scale.

Table 6. Five-fold cross-validation errors.

Model	CV-RMSE (mean ± sd)	CV-MAE (mean ± sd)
Model A	6,049.69 ± 682.65	4,160.42 ± 364.61
Model B	4,415.54 ± 689.56	2,380.68 ± 240.95
Model Bstep	4,437.75 ± 702.79	2,389.85 ± 234.37

The CV-RMSE for Model B (4,415.54) differs from the single-split test RMSE (4,351.99) by only 1.4%, well within the fold-level sd of 689.56. This confirms that the 80/20 split in §3.1 was representative, validating the test-set comparisons made in earlier sections. More importantly, the B versus B_step CV-RMSE gap of 22.21 (0.5%) is dwarfed by the same fold-level sd, resolving the partial F ambiguity from §3.4: the two models are indistinguishable in predictive performance, so the choice between them reduces to parsimony, addressed in §4.

3.6 Region ANOVA and Multiple Comparison

Research Question 2 asks whether regional differences in charges are real or merely an artifact of where high-risk individuals happen to live. We now test the region effect at two levels: marginally, ignoring other predictors, and conditionally, after controlling for the full Model B specification.

Marginal test. A one-way ANOVA on charges ~ region yields F = 1.674 on (3, 1066) df, p = 0.171, no detectable difference. Pairwise t-tests with Bonferroni correction give adjusted p-values

ranging from 0.24 to 1.00 across all six region pairs, none reaching significance. Taken at face value, region appears irrelevant to the charges.

Controlled test. We next test the joint significance of the three region dummies in Model B, which controls for age, BMI, smoker, the smoker × BMI and smoker × obese interactions, children, and sex. The joint F-test rejects the null that all three region coefficients are zero: F = 3.99 on (3, 1058) df, p = 0.0077. The estimated coefficients show a clear geographic pattern, with the Northeast as the most expensive region:

Table 7. Adjusted regional effects from Model B, using Northeast as the reference region.

Region	Estimate (vs NE)	95% CI	Significant?
Northwest	−$521.60	[−$1,297.83, $254.64]	No
Southeast	−$1,112.84	[−$1,887.31, −$338.38]	Yes
Southwest	−$1,187.81	[−$1,951.24, −$424.39]	Yes

Reconciling the reversal. The marginal test suggests that region does not matter, but the controlled test shows a significant adjusted region effect. The adjusted coefficients indicate that Northeast has the highest predicted charges, while Southeast and Southwest are significantly lower than Northeast after controlling for age, BMI, smoking status, obesity, children, and sex. Southeast provides a clear example of the confounding pattern: it has the highest smoking rate and mean BMI in the sample, which raises its raw average charges and can hide its lower adjusted coefficient. Thus, the marginal non-significance does not imply that region has no adjusted association with charges; rather, raw regional averages are partly distorted by differences in risk composition across regions.

3.7 Age Linearity Check

Purpose. Research Question 3 asks whether age affects charges linearly or whether costs accelerate at certain life stages. Figure 1 in §2.2 showed a broadly linear age-charges relationship with mild upward curvature, and §2.3 deferred a formal test to this section.

Testing strategy. We compare linear age against three nonlinear alternatives via partial F-tests, then run a lack-of-fit test against the fully saturated factor(age) model. The three partial F-tests all reject linearity (p < 1e-4), but the lack-of-fit test does not (F = 1.35, p = 0.065): the curvature flagged by the partial tests is too small to register against the saturated model.

Table 8. Tests of nonlinear age specifications and test-set RMSE comparison.

Test	F	p	Test RMSE
Model B (linear age)	—	—	4,352.0
+ poly(age, 2)	18.81	1.58e-5	4,306.5
+ poly(age, 3)	9.55	7.79e-5	4,308.1
+ ns(age, df = 4)	7.81	3.70e-5	4,298.8

Lack-of-fit (vs full age factor)	1.35	0.065	—

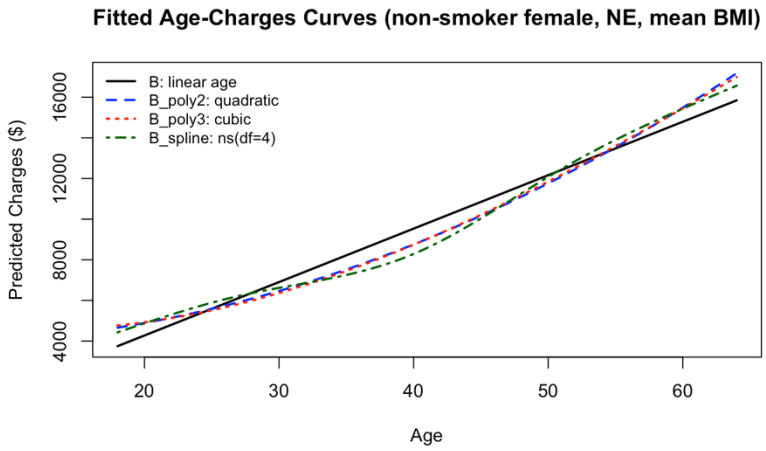


Figure 7: Predicted charges versus age across four specifications

Predictive impact. The best nonlinear alternative reduces test RMSE by only \$53 (1.22%), well within the cross-validation noise of ±\$690. Figure 7 confirms visually: the four curves are nearly indistinguishable across 18-64. The detected curvature is real but too small to matter.

Decision. We retain linear age. The trade-off costs under 1.3% in test RMSE but preserves a directly interpretable coefficient: each additional year of age is associated with approximately \$263 in additional annual charges (95% CI: [\$244, \$282]), a single number more useful for actuarial reporting than a spline basis.

4 Results and Discussion

4.1 Final Model Specification

We adopt Model B (12 parameters, linear age, interactions included) as our final model:

$$\text{charges} = \beta_0 + \beta_1 \text{age} + \beta_2 \text{bmi} + \beta_3 \text{children} + \beta_4 \text{sex} + \beta_5 \text{smoker} + \beta_6 \text{region} + \beta_7 \text{obese} + \beta_8 (\text{bmi} \times \text{smoker}) + \beta_9 (\text{smoker} \times \text{obese}) + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

4.2 Substantive Findings from the Final Model

The final model shows that medical insurance charges are not driven by any single predictor alone. Instead, the data reveal a clear interaction structure. The most important finding is that the effect of BMI on charges depends heavily on smoking status. For non-smokers, BMI has no detectable association with charges. For smokers, however, each additional BMI unit is

associated with a substantial increase in predicted annual charges. In addition, when a smoker's BMI reaches or exceeds 30, entering the obese category, predicted charges increase by roughly an additional $15,000. Therefore, the main driver of high medical costs is not BMI by itself, but the combination of smoking and high BMI or obesity.

Second, the model shows that region appears unimportant in the raw data, but has a detectable effect after controlling for individual risk factors. When we simply compare average charges across the four regions, the regional differences are not significant. However, after controlling for age, BMI, smoking status, obesity, children, sex, and the interaction terms, the Northeast has the highest adjusted predicted charges, while the Southeast and Southwest are significantly lower than the Northeast. This suggests that regional differences are not entirely absent; rather, they are partly masked by differences in risk composition across regions. However, the predictive contribution of region is small, so region is better treated as an explanatory finding rather than a major predictive driver.

Third, the model shows that a linear specification for age is sufficient. Although quadratic, cubic, and natural spline specifications detect some age nonlinearity, their improvement in test RMSE is very small. The natural spline performs best among the nonlinear alternatives, but it reduces RMSE by only about $53 compared with linear age. This gain is too small to justify the loss of interpretability. Therefore, the final model retains linear age, implying that medical charges increase steadily as age increases.

4.3 Predictive Performance Summary

Table 9 collects the test-set performance of every candidate model considered in §3.

Table 9. Final comparison of all candidate models on the held-out test set (n = 268).

Model	Parameters	Train R^2	Test RMSE ($)	Test MAE ($)
Model A	9	0.7555	6,088.63	4,221.07
Model B	12	0.8692	4,351.99	2,301.68
Model B_step	8	0.8673	4,380.63	2,287.38
Model C (Box-Cox)	12	0.8274	5,465.81	3,265.88

Note. Model C's R^2 is computed on the Box-Cox-transformed scale and is not directly comparable to the dollar-scale R^2 values for Models A, B, and B_step.

Two patterns are worth noting. First, Model B and Model B_step are essentially tied: under $30 apart on RMSE and under $15 apart on MAE. Confirming the §3.5 cross-validation conclusion that the choice between them is a parsimony judgement rather than a predictive one. Second, Model C performs materially worse than Model B on the original dollar scale, despite its diagnostically attractive transformation, confirming §3.3's decision to retain Model B.

4.4 Three Methodological Insights

Three lessons from Section 3 shaped the final modeling decision.

First, what first looked like a variance problem was mainly a structure problem. Model A had strong heteroscedasticity, with a Breusch-Pagan p-value of 5.64×10^{-19}, which could suggest using a transformation or weighted least squares. However, after adding the smoker × BMI and smoker × obese interactions, the BP p-value rose to 0.91. The issue was not unequal variance itself; the baseline model had missed an important interaction. This is why we checked the model structure before applying a technical fix.

Second, a diagnostic suggestion does not guarantee a better model. Box-Cox selected $\hat{\lambda} = 0.33$, and Model B still failed the Shapiro-Wilk normality test, so Model C looked reasonable from a diagnostic perspective. However, on the held-out test set, Model C had about 26% worse RMSE than Model B. This shows that diagnostics should be treated as clues, not final answers. A fix only matters if it improves prediction on the original dollar scale.

Third, statistical significance is not the same as practical importance. With 1,070 training observations, even small effects can become significant. The BIC-reduced model was statistically different from Model B, but CV-RMSE changed by only about 0.5%. The nonlinear age models were also significant, but the best one improved test RMSE by only about 1.2%. Therefore, the final choice was based on effect size, prediction, and interpretability, not p-values alone.

Overall, Model B was chosen because it captures the main data structure without adding unnecessary complexity.

5. Conclusion

This report examined how individual characteristics explain variation in annual medical insurance charges. Starting from a baseline additive model, we found that the most important structure in the data is the interaction between smoking status and BMI. After adding the smoker × BMI and smoker × obese interactions, Model B substantially improved predictive accuracy and resolved the main heteroscedasticity problem observed in Model A.

The main conclusions are:

- **Smoking status changes the effect of BMI on medical charges.** Among non-smokers, BMI has no detectable association with charges. Among smokers, however, higher BMI is strongly associated with higher predicted charges, and crossing the BMI = 30 obesity threshold adds roughly $15,000 in predicted annual charges.

- **The smoker-obese combination is the dominant driver of high predicted insurance charges in this dataset.** High costs are driven less by BMI alone and more by the interaction between smoking and obesity.
- **Region matters only after adjustment.** Raw regional differences are not statistically significant, but after controlling for age, BMI, smoking status, obesity, children, sex, and interaction terms, the Northeast has the highest adjusted predicted charges, while the Southeast and Southwest are significantly lower. However, region contributes only modestly to prediction.
- **Age is adequately modeled as linear.** Although nonlinear age terms are statistically significant, their test-set improvement is very small. Therefore, the linear age specification is preferred because it preserves interpretability without sacrificing meaningful predictive accuracy.

Overall, Model B is selected as the final model because it offers the best balance of predictive performance, interpretability, and connection to the research questions. It captures the main structure in the data without relying on unnecessary transformations or overly complex nonlinear terms.